

ASSIGNMENT: 2

Diagnosing Hardware Failures Theory

Task (1): List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Ans.

- 1.No display on monitor:** Blank screen, no POST beep, monitor shows "No Signal."
- 2.Faulty RAM:** PC beeps repeatedly during startup, random crashes, or fails to boot.
- 3.Hard drive/SSD failure:** Operating system does not load, "Boot Device Not Found" error, unusual clicking noises (HDD).
- 4.Overheating CPU:** PC shuts down unexpectedly, freezes, restarts, or runs very slowly with loud fan noise.

5.Power supply (PSU) failure: PC does not turn on, no lights or fan activity, or powers off suddenly during use.

Task (2): Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Ans.

1. Check the power outlet:

- Plug another device into the same outlet.
- Check: Make sure the outlet is supplying power.

2.Inspect the charger and cable:

- Look for cuts, exposed wires, bent pins, or damaged connectors.
- Check: The adapter and cable should be physically intact.

3. Check the charging connection:

- Disconnect and firmly reconnect the charger to the laptop.
- Check: Make sure the charging plug fits securely and the charging LED turns on, if present.

4. Check the charger adapter:

- Feel whether the adapter is getting unusually hot or remains completely cold.
- If available, use a multimeter to check the charger's output voltage against the voltage printed on its label.

5. Inspect the laptop charging port:

- Use a flashlight to look inside the charging port.
- Check: Look for dust, bent pins, looseness, or physical damage. Do not insert metal objects into the port.

6. Check the battery:

- If the laptop has a removable battery, remove it and try powering the laptop using only the charger.
- Check: If it works without the battery, the battery may be faulty.

7. Try a known-good compatible charger

- If available, test with another charger having the correct voltage, connector, and sufficient wattage.
- Check: If the laptop charges with the replacement charger, the original charger is likely defective.

Task(3): Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Ans.

Expected USB output:- 4.75 V – 5.25 V DC

Measured voltage:- 5.08 V DC

Conclusion:- Within the expected range - charger is working normally.

Task (4): A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Ans.

Multimeter Test of PSU 12V Rail:-

❖ Steps:

1. Shut down the PC and unplug it from the wall.
2. Disconnect the ATX power connector from the motherboard.
3. Set the multimeter to DC voltage (V $\rightleftharpoons$).
4. Identify the yellow wire (+12V) and black wire (GND).
5. Place the red probe on the yellow wire and the black probe on the black wire.
6. Turn on the PSU as appropriate for the test setup and read the multimeter.
7. Record the voltage.

Measured voltage: 11.2 V DC

Result: 11.2 V is below 11.5 V, so the 12V rail is too low.
